

Subject : Advance Java Programming

Lab Questions

Guidelines to be followed

- Develop all the applications in IntelliJ IDE
- Push it to the git repository and make it public and include it in document initial pages
- Print the document with all answers
- Share soft copy in group
- Submission Date : 25th of Chaitra,2082

Unit 1: Core Java

1. Create a **custom exception class** and demonstrate exception handling in a real scenario.
2. Implement file handling to **serialize and deserialize objects** (e.g., Student class).
3. Write a java program to show all the concepts of Object Oriented Programming
 - a. Inheritance
 - b. Abstraction
 - c. Encapsulation
 - d. Polymorphism

Unit 2: Swing GUI

1. Build an application in swing to send email with attachment
2. The application should have a form to enter details and a button to send email.
3. The email sent should be stored in the database and we should be able to see the list of sent email in swing UI.

Unit 3: Event Handling

1. Implement a GUI where **button clicks update UI dynamically**.
2. Build a **mouse event tracker** (display coordinates on click).

Unit 4: JDBC

1. Write a Java console program to perform **CRUD operations using JDBC** for a contact directory

- a. Name
- b. Email
- c. Mobile Number
- d. Conditions
 - i. No duplicate allowed in email and mobile number

Unit 5: Network Programming

1. Create a **client-server chat app using TCP sockets**.
2. Fetch data from a URL using **URLConnection**.

Unit 6: JavaFX

1. Build a **JavaFX form with validation**.
2. Create a **file chooser-based application**.

Unit 7: Servlet & JSP

1. Create a **login system using Servlet + JDBC**.
2. Develop a **JSP form processing application**.
3. Implement **session management using cookies**.

Unit 8: RMI & CORBA

1. Build a simple **RMI calculator application**.
2. Describe the architecture of RMI and CORBA with a diagram.